



Study of the industrial production index for the manufacturing of food, beverages, and tobacco products

Linear Time Series Project

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Part I : The Data

1. What does the chosen series represent ? (sector, potential data processing, logarithmic transformation, etc.)

We study the series of the unadjusted industrial production index for the manufacturing of food, beverages, and tobacco products, adjusted for seasonal variations (CVS) and working days (CJO), with a base of 100 in 2021 (Identifier 010768277). This is a monthly series containing 423 observations, spanning from January 1990 to March 2025.

The series, represented in Figure 1, begins in 1990 at a relatively high level (91) before experiencing a slight decline in the early years of the decade. This moderate contraction can be interpreted as a structural rigidity in a context of post-1980s industrial slowdown, where France's macroeconomic priority was competitive disinflation. Production then recovers from 1995 onwards, initiating a phase of cyclical expansion that leads to successive peaks in the late 1990s and early 2000s. During this period, tobacco consumption remained relatively strong, although the first control measures were already being implemented. The most significant was the **Évin law**, passed in January 1991, which banned all advertising for tobacco and alcohol, made health warnings on cigarette packs mandatory, and imposed restrictions on sales to minors. This law, embedded in a public health logic, marked a turning point in the regulation of tobacco products, although its economic effects on production remained limited in the short term.

From the mid-2000s, the series appears to reach a plateau, stabilizing around 100. This inflection corresponds to the progressive tightening of regulations, culminating in the **decree no. 2006-1386 of November 15, 2006**, which banned smoking in all enclosed public spaces, including restaurants, cafes, and nightclubs, with implementation from February 2007. The goal was clearly to reduce exposure to passive smoking and to discourage consumption by making tobacco products less visible in public spaces. The 2008 global financial crisis triggered a temporary drop in the index, followed by a rapid rebound. In July 2011, production reached its historical maximum (103.2), potentially driven by cyclical catch-up effects or a temporary surge in demand in the food segment. However, this recovery phase was short-lived. From 2012 onwards, the series entered a phase of stagnation, explained by several structural shifts : the rise of short supply chains, the gradual disaffection with industrial food products, and the growing social rejection of tobacco.

This rejection is institutionally embodied in a series of ambitious reforms under the national tobacco control plans. The most emblematic measure is the introduction of the **plain cigarette pack**, mandatory since **January 2017**, which removes brand logos and standardizes the appearance of packs. Combined with successive increases in tobacco taxes (notably under the French social security financing laws of 2018 and 2019), this reform aims to reduce the attractiveness of tobacco products, particularly among young people.

The most significant decline in the series is observed in **March 2020**, at the time of the first COVID-19 lockdown. The sudden halt in many production chains, logistical disruptions, and the closure of out-of-home consumption venues (cafes, bars, catering) explain a sharp, temporary collapse in activity. While production gradually recovered thereafter, the series did not regain its pre-crisis levels. Finally, the most recent years confirm a trend of erosion in the level of production. The 2023 EROPP survey¹ indicates a decline in regular smoking, particularly among 18–24-year-olds, in parallel with the spread of alternatives such as vaping. This transformation of behaviors, supported by public health policies, is anchoring the sector in a trajectory of relative decline.

Log transformation of the series

In order to stabilize the variance and facilitate economic interpretation, we apply a logarithmic transformation to the series :

$$Y_t = \log(X_t)$$

This transformation is justified since all values are strictly positive. It notably allows the first differences of the series to be interpreted as monthly percentage growth rates. This transformed series, shown in Figure 2, will be used in the subsequent

1. Survey on Representations, Opinions, and Perceptions of Psychoactive Substances conducted by the French Monitoring Centre for Drugs and Drug Addiction

stages of processing (stationarity tests, ARMA modeling, etc.).

2. Transform the series to make it stationary if necessary (differentiate it, correct the deterministic trend, etc.). Thoroughly justify your choices.

Correction for seasonal variations

Although the industrial production index series used in this study is supposed to be adjusted for seasonal variations (CVS-CJO) by INSEE, it is still relevant to formally test for the presence of a seasonal structure. To do so, we use **the Kruskal–Wallis test**, a non-parametric test that assesses whether the monthly distributions of the series differ significantly from one another. The test is based on the following null hypothesis :

H_0 :	The twelve months of the year come from the same distribution : no seasonality
H_1 :	At least one month has a different distribution : presence of seasonality

The test is applied to the log-transformed series using a categorical variable representing the month of each observation. The table below presents the results obtained :

Test statistic χ^2	Degrees of freedom	p-value	Decision
1.08	11	0.9999	Do not reject H_0

With a p-value of 0.9999, well above the 5% significance level, **we do not reject the null hypothesis**. In other words, there is no significant difference between the monthly distributions of the series, which suggests the absence of seasonality in the statistical sense.

We therefore consider that seasonality is statistically non-significant, and no automatic seasonal adjustment procedure will be applied in the subsequent processing steps.

Stationarity analysis

Before applying an ARIMA model, it is necessary to verify whether the series under study is stationary. For this purpose, we use two complementary tests : **the Augmented Dickey–Fuller (ADF) test and the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test**. These two tests have opposing null hypotheses, which allows us to cross-check the conclusions.

Test	Null hypothesis (H_0)	Statistic	p-value	Decision
ADF	The series has a unit root (non-stationary)	−2.8147	0.2333	Do not reject H_0
KPSS	The series is stationary (around a trend)	4.6648	< 0.01	Reject H_0

TABLE 1 – Stationarity tests results on the log-transformed series

The ADF test does not allow us to reject the null hypothesis of non-stationarity, while the KPSS test rejects the null hypothesis of stationarity. These two results converge : the log-transformed series at level is not stationary. We therefore conclude that a differencing of the series is necessary to proceed with the modeling.

Stationarity after differencing

As the log-transformed series was found to be non-stationary, we proceed with a first-order differencing. To address this issue, we apply a **first-order differencing**, which means we consider the transformed series defined by

$$Z_t = Y_t - Y_{t-1} = \log(X_t) - \log(X_{t-1}) = \log\left(\frac{X_t}{X_{t-1}}\right)$$

We apply the two complementary stationarity tests, ADF and KPSS, to this new series. The table below summarizes the results obtained.

Test	Null hypothesis (H_0)	Statistic	p-value	Decision
ADF	The series has a unit root (non-stationary)	-11.171	< 0.01	Reject H_0
KPSS	The series is stationary	0.0686	> 0.1	Do not reject H_0

TABLE 2 – Stationarity tests on the log-differenced series

The two tests lead to the same conclusion : the transformed series Z_t is thus stationary. The stationarity test results (see previous table) show that the logarithmic series $Y_t = \log(X_t)$ is not strictly stationary. We then say that Y_t is an integrated series of order 1, denoted as $I(1)$, meaning it becomes stationary after a first-order differencing.

We can therefore proceed with the analysis using an ARMA model on this transformed version of the series.

3. Graphically represent the chosen series before and after transforming it.

The figure in Annex 3 shows in the upper graph the raw series X_t , while the lower graph depicts its transformed version $Z_t = \log(X_t) - \log(X_{t-1})$, that is, the first-differenced logarithm.

After the transformation, the differenced series exhibits fluctuations centered around a zero mean, with no apparent trend, visually suggesting stationarity. The shocks become more visible : notably, the breaks in 2008 (financial crisis) and 2020 (health crisis) are clearly identifiable. This transformation removes trend effects and residual seasonality, making the series suitable for ARMA modeling.

Part II : ARMA Models

4. Pick (and justify your choice) an ARMA(p,q) model for your corrected time series X_t . Estimate the model parameters and check its validity.

2.1. Autocorrelation (ACF) and Partial Autocorrelation (PACF)

To model the corrected and stationary time series obtained after transformation (differencing of the logarithm of the raw series), we conducted an analysis of the autocorrelation functions (ACF and PACF). The ACF captures the correlation between Z_t and Z_{t-k} , while the PACF captures the direct effect of Z_{t-k} on Z_t , independently of intermediate lags.

Correlation analysis. The ACF representation (see Figure 4 in the annex) shows a sharp cutoff after the first lag, which is characteristic of a moving average process of order 1 (MA(1)). Conversely, the PACF shows a gradual decay, typical of an autoregressive (AR) process.

These observations support the choice of a mixed ARMA(1,1) model that combines both dynamics. However, in order not to restrict the candidate model space a priori, we also considered ARMA(p, q) models with $p \leq 4$ and $q \leq 2$. This choice is motivated by the asymmetry observed in the ACF and PACF functions : the ACF decays slowly, while the PACF cuts off more sharply after the second lag. This suggests the possibility of a higher-order AR process, which justifies exploring larger values for p . On the other hand, since the ACF drops sharply after 1 or 2 lags, a moderate number of MA terms ($q \leq 2$) is sufficient.

2.2 Model selection and estimation for ARMA(p,q)

To model the corrected and stationary series $\{Z_t\}$ (differenced logarithmic series), we considered ARMA(p, q) models for $p \in \{0, \dots, 4\}$ and $q \in \{0, \dots, 2\}$.

Comparison of models using information criteria. We estimated, using the `Arma()` function in R (with `include.mean = FALSE`), a set of ARMA(p, q) models and compared their performance according to the AIC and BIC information criteria.

ARMA Model	AIC	BIC	Ljung-Box p-value
ARMA(1,1)	-2320.678	-2308.543	0.232
ARMA(2,1)	-2319.464	-2303.284	0.213
ARMA(1,2)	-2319.318	-2303.138	0.201
ARMA(0,2)	-2318.362	-2306.114	0.078
ARMA(3,1)	-2318.249	-2306.114	0.078
ARMA(3,2)	-2317.574	-2297.563	0.155
ARMA(2,2)	-2317.503	-2297.503	0.236
ARMA(4,2)	-2316.512	-2298.707	0.170
ARMA(4,1)	-2316.033	-2291.763	0.147

TABLE 3 – ARMA model selection according to AIC, BIC and Ljung-Box test

Information criteria : AIC and BIC

The AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) allow the comparison of several models based on their goodness of fit, penalized by their complexity. They are respectively defined as $AIC = -2 \log(\hat{L}) + 2k$, $BIC = -2 \log(\hat{L}) + k \log(n)$ where $\hat{L}$ is the model's likelihood, k is the number of estimated parameters, and n is the sample size.

The table above presents the values of these criteria for all the tested models. The ARMA(1,1) model has the lowest AIC, which indicates superior fit quality while remaining parsimonious. It slightly outperforms the ARMA(2,1) and ARMA(1,2) models, which are still competitive.

Ljung-Box test on residuals

The Ljung-Box test is used to check whether the residuals of the model are autocorrelated. The hypotheses are : H_0 : the residuals are not autocorrelated (white noise) and H_1 : the residuals are autocorrelated. The test statistic is given by $Q = n(n+2) \sum_{k=1}^h \frac{\hat{\rho}_k^2}{n-k}$ where $\hat{\rho}_k$ is the autocorrelation of the residuals at lag k and h is the number of considered lags.

The ARMA(1,1) model has a p-value of 0.2324, which is well above the 5% threshold. We therefore do not reject the null hypothesis : the residuals are consistent with white noise. This validates the quality of the selected model. Some models, such as ARMA(4,0), show very low p-values, indicating autocorrelated residuals.

Test of zero-mean residuals. We also checked that the residuals of the selected model are centered. For this purpose, we performed a Student's t-test on the mean of the residuals. The hypotheses are $H_0 : E[\varepsilon_t] = 0$ (centered residuals), $H_1 : E[\varepsilon_t] \neq 0$. The estimated residual mean is slightly negative $\hat{\mu}_\varepsilon = -0.00073$, but statistically insignificant, as indicated by the p-value of 0.327 and the confidence interval containing zero. The p-value is well above the 5% threshold, so we do not reject the null hypothesis. We conclude that the residuals are compatible with a zero mean.

The 95% confidence interval for the mean of the residuals is : $[-0.00220; 0.00074]$.

Final model selection. The ARMA(1,1) model minimizes the AIC criterion and has a Ljung-Box p-value of 0.232, indicating that the residuals are not autocorrelated. This model is therefore the selected candidate.

Estimation of the ARMA(1,1) model parameters

The parameters of the ARMA(1,1) model were estimated by maximum likelihood on the differenced logarithmic series $\{Z_t\}$, using the `Arma()` function with `include.mean = FALSE`. The retained model is thus :

$$Z_t = \phi_1 Z_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$$

where ε_t is a white noise.

The parameter estimates are as follows :

Parameter	Estimate	Standard error
ϕ_1 (AR(1))	0.2747	0.0726
θ_1 (MA(1))	-0.7889	0.0466
σ^2	0.0002369	–

TABLE 4 – Estimation of the ARMA(1,1) model parameters

Significance of the coefficients : The estimated coefficients of the ARMA(1,1) model are statistically significant. For each of them, we test the null hypothesis that the coefficient is zero ($H_0 : \beta = 0$) using a Student's t-test :

$$t = \frac{\hat{\beta}}{\text{s.e.}(\hat{\beta})}. \text{ We thus obtain for the AR coefficient : } t_{\text{ar1}} = \frac{0.2747}{0.0726} \approx 3.78 \text{ and for the MA coefficient : } t_{\text{ma1}} = \frac{-0.7889}{0.0664} \approx -11.88.$$

These values are largely above (in absolute value) the usual critical thresholds (1.96 for a two-sided test at the 5% level), which leads us to reject the null hypothesis. We therefore conclude that both coefficients are significantly different from zero.

5. Write the ARIMA(p,d,q) model for the chosen series. The original series X_t having been judged non-stationary, we first applied a logarithm, then a first difference to obtain the following transformed series :

$$Z_t = \log(X_t) - \log(X_{t-1}) = \log\left(\frac{X_t}{X_{t-1}}\right)$$

The results of the stationarity tests (ADF and KPSS) confirmed that this series (Z_t) is stationary. We can thus directly model Z_t using an ARMA process.

The selected model, chosen based on the AIC and BIC criteria as well as residual tests, is an ARMA(1,1). It is written as :

$$Z_t = 0.2747 Z_{t-1} + \varepsilon_t - 0.7889 \varepsilon_{t-1} \quad \text{with } \varepsilon_t \sim \mathcal{N}(0, \sigma^2) \text{ and } \sigma^2 = 0.0002369.$$

This model is equivalent to an **ARIMA(1,1,1)** on the series X_t , since Z_t is obtained after a first differencing of $\log(X_t)$, i.e. $d = 1$.

Part III : Prediction

Let T be the length of the series. Assume the series' residuals are Gaussian.

6. Write the equation verified by the confidence region of level α on the future values (X_{T+1}, X_{T+2}) .

We consider the model estimated on the log-differenced series $Z_t = \log(X_t) - \log(X_{t-1})$ given by :

$$Z_t = \phi Z_{t-1} + \varepsilon_t - \theta \varepsilon_{t-1}, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma^2)$$

with $\phi = 0.2747$, $\theta = 0.7889$, and $\sigma^2 = 0.0002369$.

We seek to forecast the next two values of the series, (Z_{T+1}, Z_{T+2}) , based on the information available at time T . Under the assumption that the residuals ε_t are i.i.d. Gaussian, we have :

$$\begin{aligned} Z_{T+1} &= \phi Z_T + \varepsilon_{T+1} - \theta \varepsilon_T \\ Z_{T+2} &= \phi Z_{T+1} + \varepsilon_{T+2} - \theta \varepsilon_{T+1} \end{aligned}$$

By substituting the expression of Z_{T+1} into that of Z_{T+2} , we get :

$$Z_{T+2} = \phi^2 Z_T + (\phi - \theta) \varepsilon_{T+1} - \phi \theta \varepsilon_T + \varepsilon_{T+2}$$

Conditional forecasts

The forecasts of the next two values conditional on the information available at time T are :

$$\begin{aligned}\widehat{Z}_{T+1|T} &= \mathbb{E}[Z_{T+1} | \mathcal{F}_T] = \phi Z_T - \theta \varepsilon_T \\ \widehat{Z}_{T+2|T} &= \mathbb{E}[Z_{T+2} | \mathcal{F}_T] = \phi^2 Z_T - \phi \theta \varepsilon_T\end{aligned}$$

The forecast errors are thus :

$$\begin{aligned}\eta_{T+1} &= Z_{T+1} - \widehat{Z}_{T+1|T} = \varepsilon_{T+1} \\ \eta_{T+2} &= Z_{T+2} - \widehat{Z}_{T+2|T} = (\phi - \theta)\varepsilon_{T+1} + \varepsilon_{T+2}\end{aligned}$$

Variance matrix of the forecast errors

The vector of errors is :

$$U = \begin{pmatrix} \eta_{T+1} \\ \eta_{T+2} \end{pmatrix} = \begin{pmatrix} \varepsilon_{T+1} \\ (1 + \phi - \theta)\varepsilon_{T+1} + \varepsilon_{T+2} \end{pmatrix}$$

Consequently, U is a Gaussian vector with zero mean, and its variance-covariance matrix is :

$$\Sigma = \begin{pmatrix} \sigma^2 & (1 + \phi - \theta)\sigma^2 \\ (1 + \phi - \theta)\sigma^2 & [1 + (1 + \phi - \theta)^2] \sigma^2 \end{pmatrix}$$

With the numerical values :

$$1 + \phi - \theta = 1 + 0.2747 - 0.7889 = 0.4858$$

$$\Sigma \approx \begin{pmatrix} 0.0002369 & 0.0001151 \\ 0.0001151 & 0.0003539 \end{pmatrix}$$

Under the assumption that the residuals are Gaussian, we have :

$$CI_{95\%}(Z_{T+1}) = \left[\widehat{Z}_{T+1|T} \pm 1.96 \times \sqrt{\sigma^2} \right]$$

$$CI_{95\%}(Z_{T+2}) = \left[\widehat{Z}_{T+2|T} \pm 1.96 \times \sqrt{(1 + (1 + \phi - \theta)^2) \sigma^2} \right]$$

7. Give the hypotheses used to get this region.

The following assumptions are required for the confidence region constructed for (Z_{T+1}, Z_{T+2}) to be valid :

1. **Normality** : The residuals ε_t are independent and identically distributed following a normal distribution $\mathcal{N}(0, \sigma^2)$.
2. **Stationarity** : The underlying ARMA(1,1) process for Z_t is stationary, which implies that the autoregressive coefficient $\phi_1 = 0.2747$ has modulus strictly less than 1.
3. **Invertibility** : The model is invertible, meaning the MA parameter satisfies $|\theta_1| < 1$. Here, $\theta_1 = -0.7889$ satisfies this condition.
4. **Known or reliably estimated parameters** : The coefficients (ϕ_1, θ_1) and the variance σ^2 are assumed known or reliably estimated (without significant bias or excessive uncertainty).
5. **Absence of residual autocorrelation** : The model correctly captures temporal dependence : the estimated residuals must behave as white noise (no significant autocorrelation), as verified by Ljung-Box tests on the residuals.

8. Graphically represent this region for $\alpha = 95\%$. Comment on it.

Figure 5 in the appendix shows the forecasts for the next two periods (*April* and *May 2025*) on the stationary series $Z_t =$

$\log(X_t) - \log(X_{t-1})$, along with their 95% confidence intervals, under the assumption that the residuals of the ARMA(1,1) model are Gaussian.

The point forecasts are :

$$\boxed{\hat{Z}_{T+1} = 0.0303 \quad (\text{April 2025}) \quad \hat{Z}_{T+2} = 0.0083 \quad (\text{May 2025}).}$$

The corresponding confidence intervals are therefore :

$$IC_{95\%}(Z_{T+1}) = \left[0.0303 \pm 1.96 \times \sqrt{\hat{\sigma}^2} \right],$$

$$IC_{95\%}(Z_{T+2}) = \left[0.0083 \pm 1.96 \times \sqrt{(1 + (1 + \hat{\phi} - \hat{\theta})^2) \hat{\sigma}^2} \right].$$

The exact numerical bounds are shown by the blue bands around the points in the figure.

We observe that the forecasts remain within the historical range of fluctuations of Z_t , and that the widening of the intervals from one month to the next naturally reflects the increasing uncertainty as the forecast horizon extends.

Confidence ellipse

The confidence ellipse (see Figure 6 in the appendix) represents the region within which the true values of (Z_{T+1}, Z_{T+2}) lie with 95% probability, assuming Gaussian residuals.

The elongated and tilted shape of the ellipse indicates a strong positive correlation between the one-step-ahead and two-step-ahead forecast errors. This is consistent with the structure of the ARMA(1,1) model, where Z_{T+2} depends on both ε_{T+2} and ε_{T+1} .

Therefore, the prediction region is not circular but oriented, meaning that the uncertainty about Z_{T+2} is partially explained by the uncertainty about Z_{T+1} .

9. Open question : Let Y_t be a stationary time series available from $t = 1$ to T . We assume that Y_{T+1} is available earlier than X_{T+1} . Under which condition(s) does this information allow you to improve the prediction of X_{T+1} ? How would you test it/them ?

If (Y_t) is a stationary time series and Y_{T+1} is available before X_{T+1} , then Y_{T+1} can improve the prediction of X_{T+1} if Y_t Granger-causes X_t . This means that Y_t provides additional information not contained in the past history of X_t .

This can be tested using a Granger causality test or by analyzing the cross-correlation function.

To formalize this case, consider the reduced-form bivariate VAR(p) model :

$$\begin{pmatrix} X_t \\ Y_t \end{pmatrix} = \sum_{k=1}^p A_k \begin{pmatrix} X_{t-k} \\ Y_{t-k} \end{pmatrix} + u_t, \quad u_t \sim \mathcal{N}(0, \Sigma), \quad \Sigma = \begin{pmatrix} \sigma_{XX} & \sigma_{XY} \\ \sigma_{XY} & \sigma_{YY} \end{pmatrix}$$

Instantaneous Granger causality from Y to X holds if :

$$H_0 : \sigma_{XY} = 0 \quad \text{vs} \quad H_1 : \sigma_{XY} \neq 0$$

This hypothesis can be tested using a Wald, likelihood ratio (LR), or Lagrange multiplier (LM) test. For example, the Wald statistic :

$$W = T \cdot \hat{\rho}_{XY}^2 = \frac{\left(\sum_{t=1}^T \hat{u}_{X_t} \hat{u}_{Y_t} \right)^2}{\sum_{t=1}^T \hat{u}_{X_t}^2 \cdot \sum_{t=1}^T \hat{u}_{Y_t}^2}$$

follows an asymptotic χ_1^2 distribution under H_0 . Rejecting H_0 implies that knowing Y_{T+1} as soon as it is available allows for a more accurate forecast of X_{T+1} .

Appendix

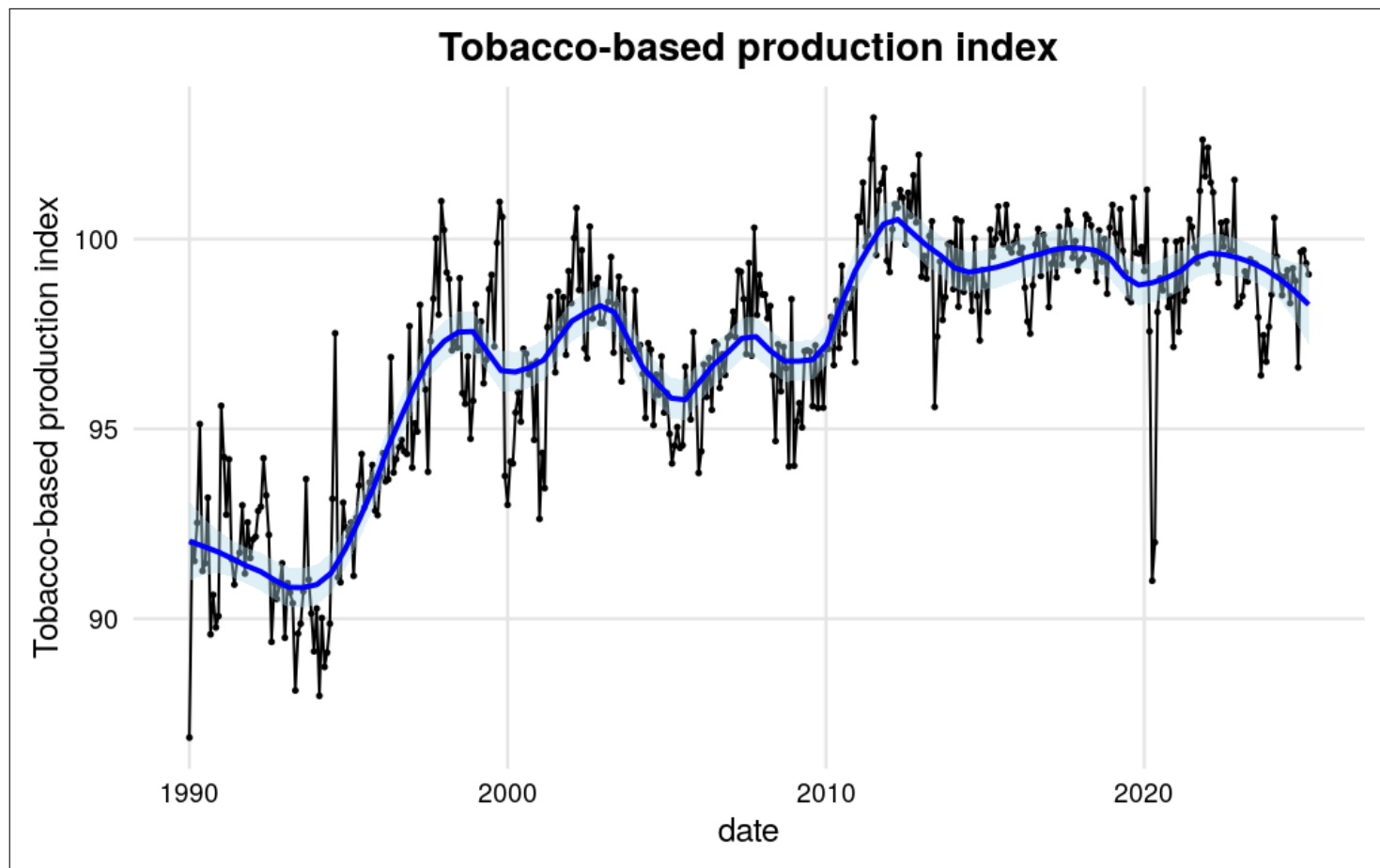


FIGURE 1 – CVS-CJO Index of industrial production (base 100 in 2021) : manufacture of food products, beverages, and tobacco products in France from January 1990 to March 2025

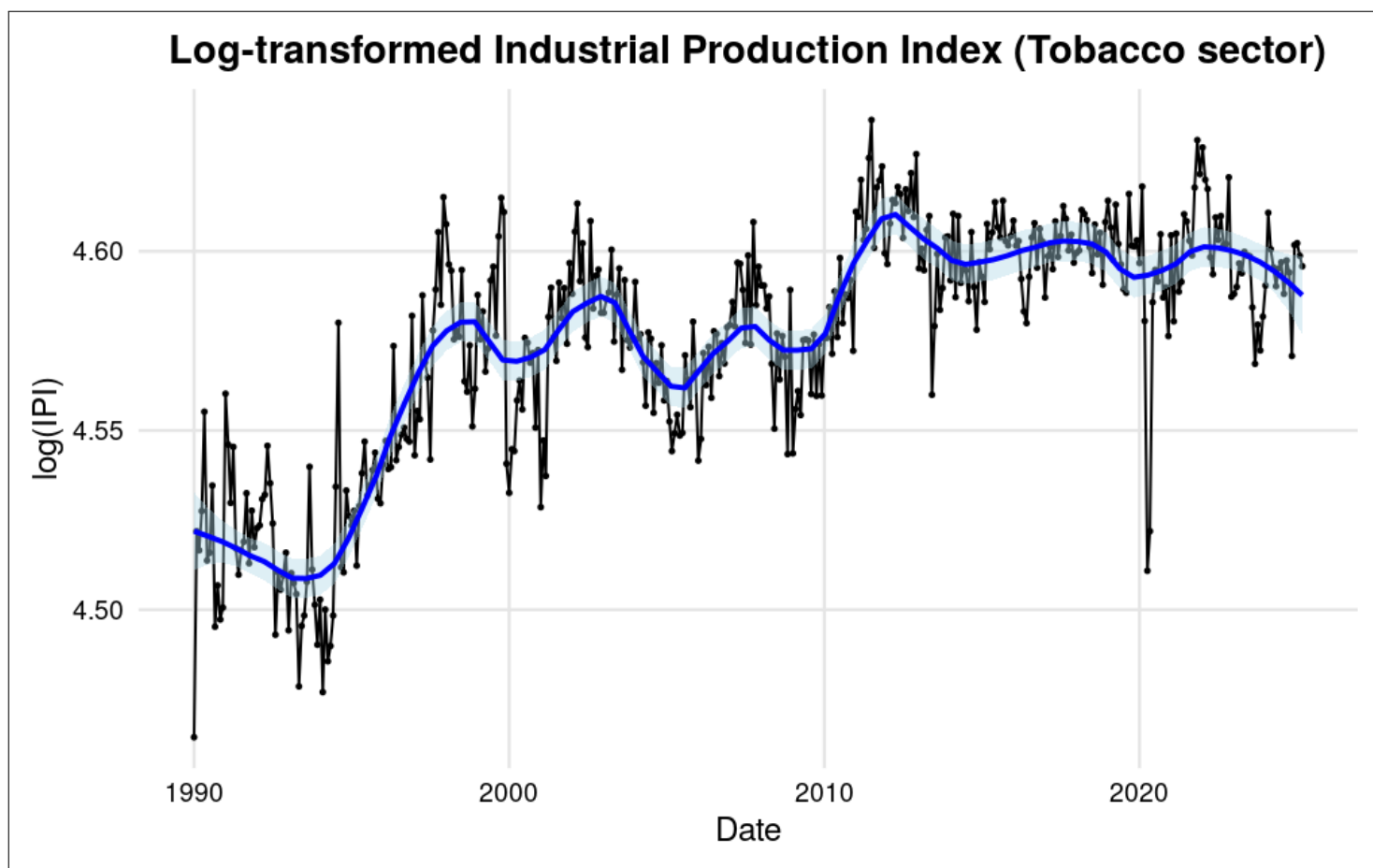


FIGURE 2 – Log-transformed CVS-CJO index of industrial production (base 100 in 2021) from January 1990 to March 2025

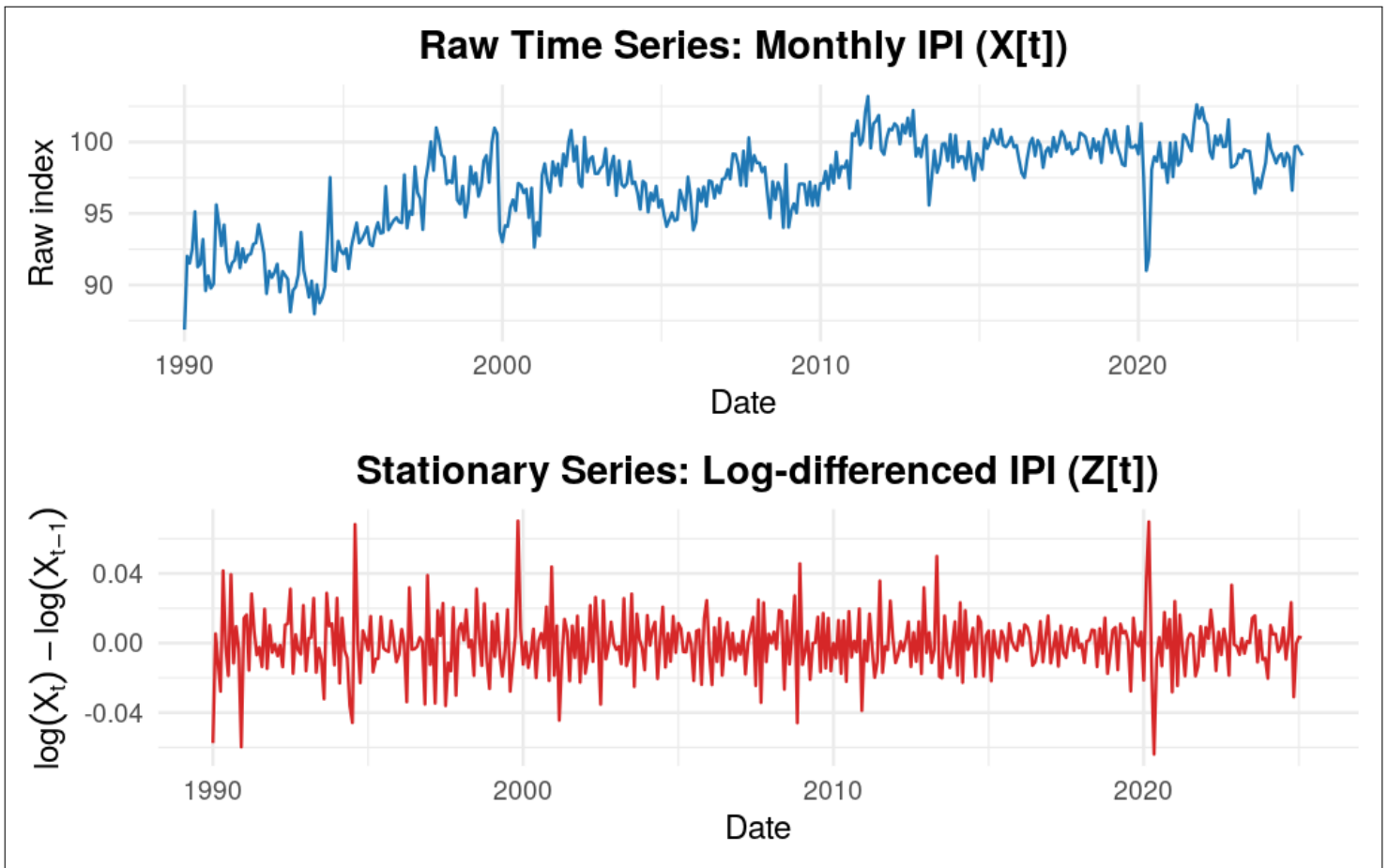


FIGURE 3 – Original raw series (left) and series after log-transformation and differencing (right)

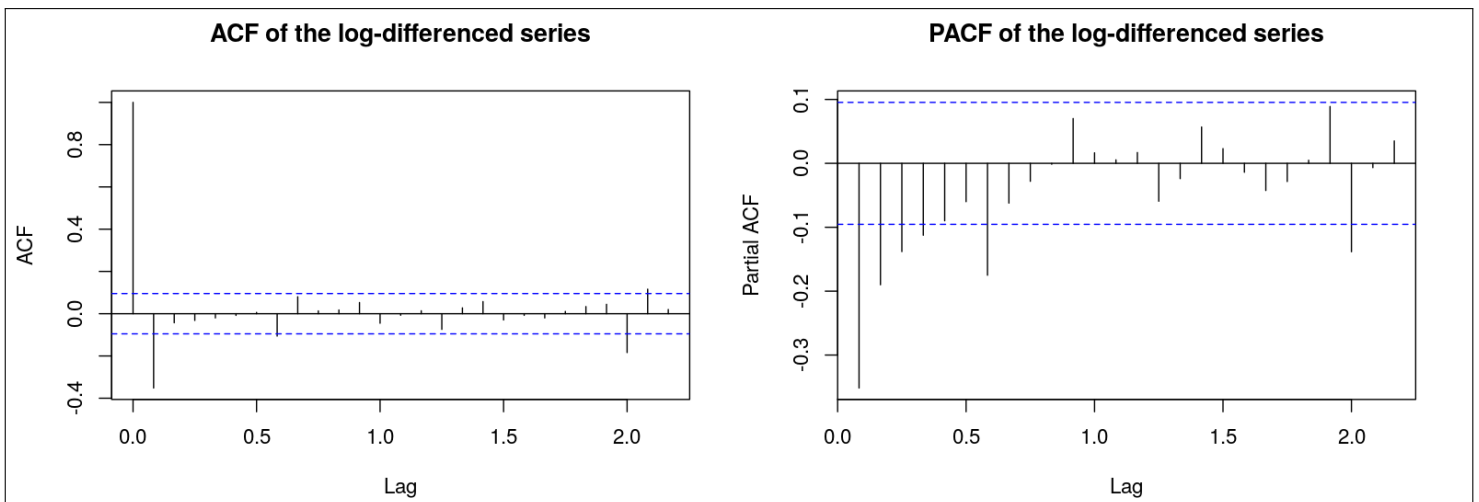


FIGURE 4 – ACF and PACF of the log-differenced series (Z_t)

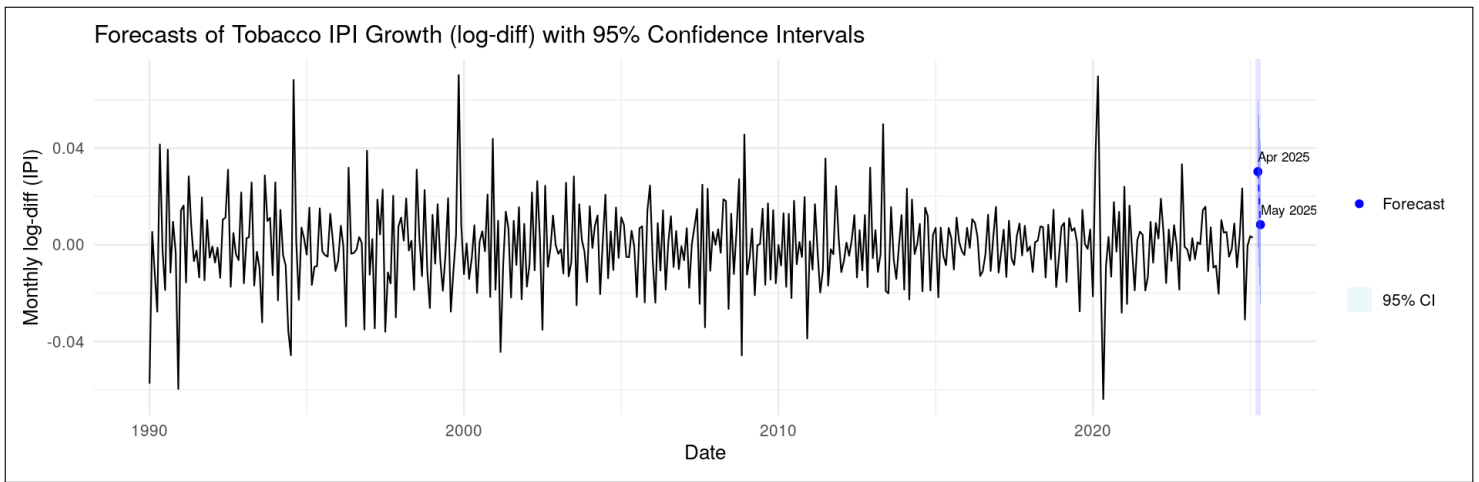


FIGURE 5 – Predictions on the series Z_t (log-differences) with 95% confidence intervals

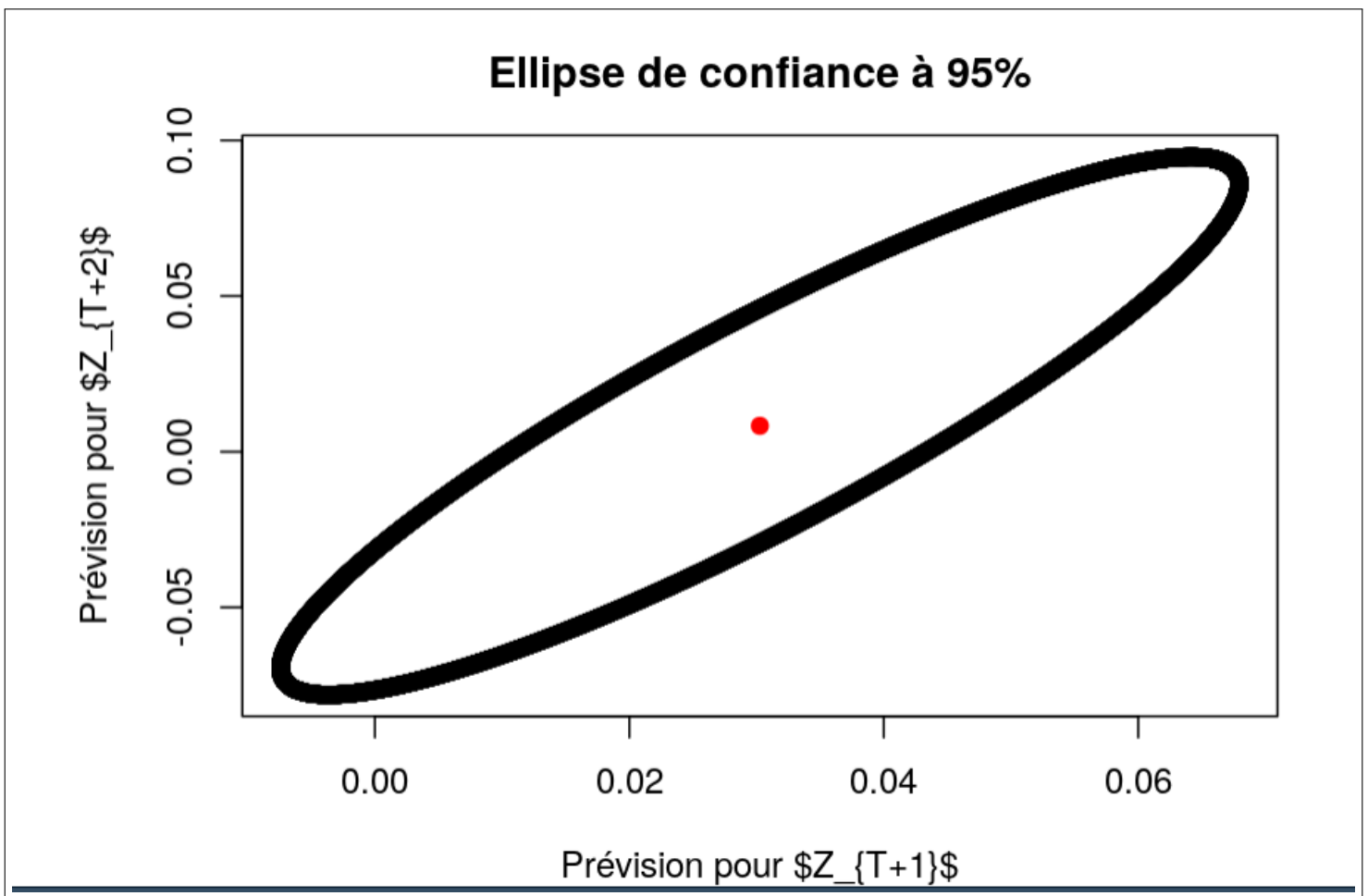


FIGURE 6 – 95% confidence ellipse for (Z_{T+1}, Z_{T+2}) under the assumption of Gaussian residuals

Appendix : R code

The following code was used for the analysis, stationarity tests, ARMA modeling, and forecasting.

```

1 # =====
2 # ANALYSE DE LA SERIE TEMPORELLE IPI (TABAC)
3 # =====
4
5 # Chargement des packages necessaires

```

```

6 library(tidyverse) # Pour manipulation de donnees et ggplot2
7 library(lubridate) # Pour gestion des dates
8 library(forecast) # Pour les mod les ARIMA et les previsions
9 library(tseries) # Pour les tests ADF
10 library(urca) # Pour les tests KPSS
11 library(gridExtra) # Pour agencer les graphiques
12 library(ellipse) # Pour tracer des ellipses de confiance
13
14 # =====
15 # QUESTION 1 : Serie brute
16 # =====
17
18 # Chargement et nettoyage des donnees
19 df <- read.csv("valeurs_mensuelles.csv", sep = ";", header = FALSE)
20
21 # On enleve les lignes d'en-tete inutiles et on renomme les colonnes
22 data_brute <- df[4:nrow(df), ]
23 colnames(data_brute) <- c("date", "value", "code")
24
25 # Transformation des colonnes et nettoyage
26 data_brute <- data_brute %>%
27   mutate(date = ymd(paste0(date, "-01")),
28          value = as.numeric(value)) %>%
29   select(date, value) %>%
30   drop_na()
31
32 # Visualisation de la serie brute
33 ggplot(data_brute, aes(x = date, y = value)) +
34   geom_line(color = "black") +
35   geom_point(size = 0.6) +
36   geom_smooth(method = "loess", span = 0.2, se = TRUE,
37              color = "blue", fill = "lightblue", linewidth = 1) +
38   labs(title = "Indice de production IPI - Produits du tabac",
39        x = "Date", y = "IPI (Tabac)") +
40   theme_minimal(base_size = 13)
41
42 # Transformation logarithmique
43 data <- data_brute %>% mutate(log_value = log(value))
44
45 # Visualisation de la serie transformee
46 ggplot(data, aes(x = date, y = log_value)) +
47   geom_line(color = "black") +
48   geom_point(size = 0.6) +
49   geom_smooth(method = "loess", span = 0.2, se = TRUE,
50              color = "blue", fill = "lightblue", linewidth = 1) +
51   labs(title = "IPI transforme en logarithme",
52        x = "Date", y = "log(IPI)") +
53   theme_minimal(base_size = 13)
54
55 # Conversion en objet ts
56 ts_log <- ts(data$log_value,
57              start = c(year(min(data$date)), month(min(data$date))),
58              frequency = 12)
59
60 # =====
61 # QUESTION 2 : Stationnarite
62 # =====
63
64 # a) Test de saisonnalite (Kruskal-Wallis)
65 data <- data %>%
66   mutate(mois = factor(month(date)), annee = year(date))

```

```

67 kruskal.test(log_value ~ mois, data = data)
68
69 # b) Tests de stationnarite
70 adf.test(data$log_value)
71 kpss.test(data$log_value)
72
73 # Transformation differenciee
74 data <- data %>%
75   mutate(log_diff = c(NA, diff(log_value)))
76 data_diff <- data %>% drop_na(log_diff)
77
78 # Tests sur serie differenciee
79 adf.test(data_diff$log_diff)
80 kpss.test(data_diff$log_diff)
81
82 # =====
83 # QUESTION 3 : Visualisations
84 # =====
85
86 data_trans <- data_brute %>%
87   mutate(log_value = log(value),
88          log_diff = c(NA, diff(log(value))))
89
90 # Serie brute
91 p1 <- ggplot(data_trans, aes(x = date, y = value)) +
92   geom_line(color = "#1f77b4", size = 0.6) +
93   labs(title = "S rie brute IPI(X_t)", x = "Date", y = "Indice") +
94   theme_minimal(base_size = 13)
95
96 # Serie differenciee
97 p2 <- ggplot(data_trans, aes(x = date, y = log_diff)) +
98   geom_line(color = "#d62728", size = 0.6) +
99   labs(title = "Diff rence logarithmique(Z_t)", x = "Date",
100        y = expression(log(X[t]) - log(X[t-1]))) +
101   theme_minimal(base_size = 13)
102
103 grid.arrange(p1, p2, ncol = 1)
104
105 # =====
106 # QUESTION 4 : ACF / PACF
107 # =====
108
109 log_diff <- na.omit(data_trans$log_diff)
110 ts_log_diff <- ts(log_diff, start = c(1990, 1), frequency = 12)
111
112 par(mfrow = c(1, 2))
113 acf(ts_log_diff, main = "ACF de Z_t")
114 pacf(ts_log_diff, main = "PACF de Z_t")
115
116 # =====
117 # QUESTION 5 : Selection ARMA(p,q)
118 # =====
119
120 results <- data.frame(model = character(), p = integer(), q = integer(),
121                      AIC = numeric(), BIC = numeric(), ljung_pval = numeric())
122
123 for (p in 0:4) {
124   for (q in 0:2) {
125     fit <- tryCatch({
126       Arima(ts_log_diff, order = c(p, 0, q), include.mean = FALSE)
127     }, error = function(e) return(NULL))

```

```

128
129   if (!is.null(fit)) {
130     lb <- Box.test(residuals(fit), lag = 10, type = "Ljung-Box", fitdf = p + q)
131     results <- rbind(results, data.frame(
132       model = paste0("ARMA(", p, ",", q, ")"),
133       p = p, q = q,
134       AIC = AIC(fit), BIC = BIC(fit), ljung_pval = lb$p.value))
135   }
136 }
137 }
138 results <- results[order(results$AIC), ]
139 print(results)
140
141 # Test de moyenne nulle des residus
142 modele_arma <- Arima(ts_log_diff, order=c(1,0,1), include.mean = FALSE)
143 residus <- residuals(modele_arma)
144 t.test(residus, mu = 0)
145 summary(modele_arma)
146
147 # QQ-plot des residus
148 qqnorm(residus, main = "QQ-plot des residus")
149 qqline(residus, col = "red", lwd = 2)
150
151 # =====
152 # QUESTION 6-7 : Previsions & Ellipse
153 # =====
154
155 Pred <- predict(modele_arma, n.ahead = 2, se.fit = TRUE)
156 phi <- as.numeric(modele_arma$coef["ar1"])
157 theta <- as.numeric(modele_arma$coef["ma1"])
158 sigma2 <- modele_arma$sigma2
159 d <- 1 + phi - theta
160
161 Sigma <- matrix(c(
162   sigma2,
163   d * sigma2,
164   d * sigma2,
165   (1 + d^2) * sigma2
166 ), nrow = 2, byrow = TRUE)
167
168 ell <- ellipse(Sigma, centre = Pred$pred, level = 0.95, npoints = 10000)
169 plot(ell, xlab = "Pr vision_Z_{T+1}", ylab = "Pr vision_Z_{T+2}", main = "Ellipse de confiance 95%")
170 points(Pred$pred[1], Pred$pred[2], pch = 19, col = "red")
171
172 # Previsions sur les dates a venir
173 last_date <- max(data$date)
174 df_pred_zt <- data.frame(
175   date = seq(last_date %m+% months(1), by = "1month", length.out = 2),
176   Zt = Pred$pred,
177   lower = Pred$pred - 1.96 * Pred$se,
178   upper = Pred$pred + 1.96 * Pred$se
179 )
180
181 df_past_zt <- data %>%
182   select(date, log_diff) %>%
183   drop_na() %>%
184   rename(Zt = log_diff)
185
186 # Visualisation finale
187 ggplot(df_past_zt, aes(x = date, y = Zt)) +
188   geom_line(color = "black") +

```

```

189 geom_ribbon(data = df_pred_zt, aes(ymin = lower, ymax = upper), fill = "blue", alpha = 0.3) +
190 geom_line(data = df_pred_zt, aes(x = date, y = Zt), color = "blue", linetype = "dashed") +
191 geom_point(data = df_pred_zt, aes(x = date, y = Zt), color = "blue", size = 2) +
192 geom_text(data = df_pred_zt, aes(x = date, y = Zt, label = c("Avril_2025", "Mai_2025")),
193           vjust = -1, hjust = 0, size = 3) +
194 annotate("rect", xmin = as.Date("2025-03-01"), xmax = as.Date("2025-05-01"),
195          ymin = -Inf, ymax = Inf, alpha = 0.1, fill = "blue") +
196 labs(title = "Pr vision du taux de croissance IPI (log-diff) avec IC 95%",
197       x = "Date", y = "log-diff mensuel (IPI)") +
198 theme_minimal(base_size = 13)

```